

PPE Compliance and Worker Safety Monitoring System

1. Solution Overview

1.1 Objective

The objective of this solution is to automatically monitor worker safety compliance from video footage by:

- Detecting and tracking workers in real time.
- Detecting Personal Protective Equipment (PPE) such as safety helmets and safety vests.
- Identifying PPE violations.
- Detecting unsafe proximity between workers and machinery/vehicles.
- Generating compliance reports and violation snapshots for audit purposes.

The system processes video streams frame-by-frame and produces both visual monitoring outputs and structured reports.

2. Detection Approach

2.1 Person Detection and Tracking

The first stage of the pipeline detects workers in each frame using a dedicated person detection model.

Workflow:

1. Video frames are read from the input source.
2. Frames are resized to reduce computational load.
3. A YOLO-based person detector identifies workers.
4. ByteTrack tracking assigns a persistent ID to each worker.
5. Worker IDs are maintained across frames to enable continuous monitoring.

Output:

- Worker bounding box
- Worker tracking ID
- Worker count per frame

Tracking enables the system to monitor individual workers over time and generate worker-specific compliance reports.

2.2 PPE Detection

A separate PPE detection model identifies safety-related objects in the scene.

Detected classes include:

- Hardhat
- Safety Vest
- Mask
- No Hardhat
- No Mask
- No Safety Vest
- Machinery
- Vehicle
- Safety Cone
- Person

For this implementation, the following classes are primarily used:

- Hardhat
- Safety Vest
- Machinery
- Vehicle

Each detected PPE item is represented by a bounding box and confidence score.

2.3 PPE Association Logic

Since PPE and workers are detected by separate models, an association step is required.

Association Method

For every detected PPE object:

1. Compute the PPE bounding box center point.
2. Check whether the center lies inside a tracked worker bounding box.
3. If the PPE center falls within the worker region:
4. Hardhat → Helmet present
5. Safety Vest → Vest present

This creates a worker-level PPE status.

Example:

Worker ID 12

- Helmet detected inside worker box → Helmet = True
- Vest detected inside worker box → Vest = True

Worker ID 15

- No helmet detected
- Vest detected

Result:

- Helmet = False
 - Vest = True
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2.4 Violation Detection

The system evaluates PPE compliance for every tracked worker.

Rules:

Condition	Violation
Helmet missing	No Helmet
Safety vest missing	No Vest

A worker can have multiple simultaneous violations.

Examples:

Helmet	Vest	Result
Yes	Yes	Compliant
No	Yes	No Helmet
Yes	No	No Vest
No	No	No Helmet, No Vest

Violation events are logged with:

- Timestamp
- Worker ID
- Violation type

Snapshots are saved automatically for evidence and reporting.

2.5 Unsafe Proximity Detection

The PPE model also detects machinery and vehicles.

For every worker:

1. Calculate worker bounding box center.
2. Calculate equipment bounding box center.
3. Compute Euclidean distance between centers.

Distance:

$$\text{Distance} = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

If the distance is below a predefined threshold:

- Unsafe Proximity event is triggered.
- Event is recorded in the report.
- Visual warning is displayed.

This provides an additional layer of safety monitoring beyond PPE compliance.

2.6 Reporting

The system automatically generates CSV reports:

Worker Count Report

Contains:

- Timestamp
- Number of workers detected

PPE Compliance Report

Contains:

- Timestamp
- Worker ID
- Helmet status
- Vest status

Violation Report

Contains:

- Timestamp
- Worker ID
- Violation type

Unsafe Proximity Report

Contains:

- Timestamp
- Worker ID
- Unsafe proximity event

Snapshots of violations are stored for future investigation and auditing.

3. Model Selection

3.1 Models Considered

Several object detection approaches were evaluated for worker and PPE monitoring.

Option 1 – Traditional Computer Vision

Examples:

- HOG + SVM
- Haar Cascades
- Background Subtraction

Advantages:

- Lightweight
- Minimal hardware requirements

Limitations:

- Poor robustness to lighting changes
- Weak performance under occlusion
- Difficult to scale for multiple PPE classes

Not selected due to insufficient accuracy for industrial environments.

Option 2 – Two-Stage Detectors

Examples:

- Faster R-CNN
- Mask R-CNN

Advantages:

- High detection accuracy
- Strong localization performance

Limitations:

- Higher inference latency
- Increased computational requirements
- Less suitable for real-time video monitoring

Not selected because real-time performance was a key requirement.

Option 3 – Single-Stage YOLO Detectors

Examples:

- YOLOv5
- YOLOv8
- YOLOv11

Advantages:

- Real-time inference
- High detection accuracy
- Easy deployment
- Strong community support

These models provide an effective balance between speed and accuracy.

3.2 Final Models Selected

Person Detection Model

Model: YOLO Person Detector (ONNX)

Purpose:

- Detect workers in each frame.

- Provide input to ByteTrack for worker tracking.

Reasons for Selection:

- Fast inference performance.
 - Optimized ONNX deployment.
 - Reliable human detection.
 - Suitable for real-time processing.
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PPE Detection Model

Model: Custom PPE YOLO Detector

Purpose:

- Detect helmets, vests, machinery, vehicles, and other safety-related objects.

Reasons for Selection:

- Pretrained on PPE-related classes.
 - Detects multiple safety objects using a single model.
 - Real-time performance.
 - Simplifies deployment by avoiding multiple specialized detectors.
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Tracking Algorithm

Algorithm: ByteTrack

Purpose:

- Maintain worker identities across frames.

Reasons for Selection:

- State-of-the-art multi-object tracking.
 - High tracking stability.
 - Effective handling of temporary missed detections.
 - Native integration with YOLO workflows.
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3.3 Final Architecture

Video Input

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Person Detection (YOLO)



ByteTrack Tracking



Worker IDs



PPE Detection (YOLO)



PPE-to-Worker Association



Compliance Evaluation



Violation & Proximity Detection



Reporting and Snapshot Generation

This architecture provides a practical balance between detection accuracy, computational efficiency, scalability, and ease of deployment for industrial safety monitoring applications.